# Empirical NTK derivation

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# What is NTK?

## 1. Notation

$$f : \mathbb{R}^{n_0} \longrightarrow \mathbb{R}^{n_L}$$

input: $n_0$ dimension

output: $n_L$ dimension

$\Theta$: $f$'s parameters $\in \mathbb{R}^p$

$p$: number of $f$'s parameters

$J$: Jacobian

$$J_x f = \left[ \nabla_x^T f_i \right]_i = \left[ \frac{\partial f_i}{\partial x_j} \right]_{ij} : n_L \times n_0$$

$$J_\theta f(x) = \left[ \nabla_\theta^T f_i(x) \right]_i = \left[ \frac{\partial f_i(x)}{\partial \theta_j} \right]_{ij} : n_L \times p$$

$$J_\theta f(x_1) \, J_\theta f(x_2)^T = \left[ \sum_{k=1}^{p} \frac{\partial f_i(x_1)}{\partial \theta_k} \cdot \frac{\partial f_j(x_2)}{\partial \theta_k} \right]_{ij} : n_L \times n_L$$

$\hookrightarrow (n_L \times p) @ (p \times n_L)$

$$\mathrm{tr}\left( J_\theta f(x_1) \, J_\theta f(x_2)^T \right) = \sum_{m=1}^{n_L} \left( \sum_{k=1}^{p} \frac{\partial f_k(x_1)}{\partial \theta_k} \cdot \frac{\partial f_k(x_2)}{\partial \theta_k} \right)$$

$\hookrightarrow n_L$ sequential $p \times p$

$$\mathrm{tr}\left( J_\theta f(x_1) J_\theta f(x_2)^T \right) = \sum_{k=1}^{n_L} \left( \sum_{m=1}^{p} \frac{\partial f_k(x_1)}{\partial \theta_m} \cdot \frac{\partial f_k(x_2)}{\partial \theta_m} \right)$$

Q: what if we compute Jacobian of sum of logits?

$$\sum_{m=1}^{p} \frac{\partial \sum_{k=1}^{n_L} f_k(x_1)}{\partial \theta_m} \cdot \frac{\partial \sum_{k=1}^{n_L} f_k(x_2)}{\partial \theta_m}$$

for a specific $m : p$.

$$\frac{\partial \sum_{k=1}^{n_L} f_k(x_1)}{\partial \theta_p} \cdot \frac{\partial \sum_{k=1}^{n_L} f_k(x_2)}{\partial \theta_p} \quad \text{vs} \quad \sum_{k=1}^{n_L} \frac{\partial f_k(x_1)}{\partial \theta_p} \cdot \frac{\partial f_k(x_2)}{\partial \theta_p}$$

$$\Longrightarrow \sum_{i=1}^{n_L} \langle z_i \cdot z_i \rangle + \sum_{i=1}^{n_L} \sum_{\substack{j=1, \\ j \neq i}}^{n_L} \langle z_i, z_j \rangle$$

zero off-diagonal

$$\frac{\partial \sum_{k=1}^{n_L} f_k(x_1)}{\partial \theta_p} \cdot \frac{\partial \sum_{k=1}^{n_L} f_k(x_2)}{\partial \theta_p} = \left(\sum_{k=1}^{n_L} z_k\right) \cdot \left(\sum_{k=1}^{n_L} z_k\right)$$

$$= \sum_{k=1}^{n_L} \langle z_k \cdot z_k \rangle + \sum_{k=1}^{n_L} \sum_{\substack{o=1 \\ o \neq k}}^{n_L} z_k z_o$$

$$\Longrightarrow \frac{\partial \sum_{k=1}^{n_L} f_k(x_1)}{\partial \theta_p} \cdot \frac{\partial \sum_{k=1}^{n_L} f_k(x_2)}{\partial \theta_p}$$

$$= \sum_{k=1}^{n_L} \frac{\partial f_k(x_1)}{\partial \theta_p} \cdot \frac{\partial f_k(x_2)}{\partial \theta_p}$$

if $\forall\, k, o \in [1, n_L];\ k \neq o$

$$\frac{\partial f_k(x_1)}{\partial \theta_p} \cdot \frac{\partial f_k(x_2)}{\partial \theta_p} = 0$$

So we can safely compute NTK with taking jacobian of sum of logits w.r.t params only if $\forall i, j;\ i \neq j \Rightarrow \frac{\partial f_i(x)}{\partial \theta} \cdot \frac{\partial f_j(x)}{\partial \theta} = 0$

which means logits are i.i.d.